3D-PRINTED QUAD-DRONE

PROJECT NAME:	3D-Printed Quad-Drone
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WHY THIS PROJECT ?

- First Person View (FPV) drones are widely used in racing, inspection, research and defense-related applications. However, learning FPV flight on high-cost commercial drones often leads to damage and financial loss during training.
- This project was undertaken to develop a low-cost and mass producible specifically intended for training and flight-testing purposes. By using a 3D-printed frame, the design can be rapidly modified, repaired, or reprinted, making it ideal for beginners and experimental testing.

INTRODUCTION

- The 3D-Printed FPV Drone is a lightweight quadcopter designed primarily to test FPV flight characteristics and to train new pilots.
- The frame is fully 3D printed, allowing design flexibility and quick iteration. The drone serves as a testbed platform for experimenting with different FPV setups, flight controller tuning, and payload arrangements.

KEY FEATURES

- 3D-Printed Airframe enables rapid prototyping, customization, and low-cost replacement.
- FPV Training Platform
- Flight Testing Platform
- Used to test FPV flight dynamics before integrating FPV into advanced drones.
- Allows easy replacement and upgrading of components.
- Cost-Effective Development
- Significantly reduces risk and expense during pilot training.

ARCHITECTURE

- Use of 3D Printing
- 3D printing was chosen to:
- Reduce manufacturing cost
- Allow fast design iterations
- Enable lightweight structural designs
- Simplify repairs by reprinting damaged parts
- The frame design can be modified based on flight performance and pilot feedback.

RESULTS

- The 3D-printed FPV drone successfully demonstrated stable flight and effective FPV operation
- Minor vibrations were observed due to the lightweight printed structure, which provided valuable data for future design improvements.

FUTURE SCOPE

1. **Frame reinforcement using composite or carbon-infused filaments**
2. **Improved vibration damping techniques**
3. **Use as a swarm or multi-drone training platform**